Abubakr M. Abdelhalim

Electrical engineer — hardware design, embedded systems, power and signal integrity

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Final-year electrical engineering student at Universiti Teknologi Malaysia, graduating February 2027. At Pythonium.ai I worked on the hardware design of an industrial DAQ module and a rigid-flex SSD tester adapter, from part selection and simulation to schematic review. Gold Medal, MIIE 2026; Industry Award, UTM EE Capstone Showcase 2026.

EXPERIENCE

Embedded / PCB Design Engineer Intern · Pythonium.ai — Jul – Oct 2026
Johor Bahru, Malaysia

- Industrial DAQ module on an STM32H563: AD4111 front end for 8 single-ended (or 4 differential) voltage inputs and 4 current inputs, 2 MAX31865 RTD inputs, AD3531 analog outputs, LT3514 power tree, LAN8742A Ethernet and RS-485.
- Stabilised the analog output stage with an isolation resistor and feedback capacitor, verified in simulation; ran schematic reviews with findings logged by severity.
- Designed a passive M.2-to-EDSFF adapter linking an SSD tester to an on-board diagnostic card, with footprints built from the connector spec dimensions.
- Reworked the adapter as a rigid-flex board designed to IPC-2223: bend radius, flex-zone routing and rigid-to-flex transitions.

Part-Time Engineer · TSH Group — Apr – Jul 2025
Factory automation and commissioning, Johor

- PLC testing and industrial wiring for automation systems; wrote the work instructions for MTC machine testing.
- Designed and built a DI/DO testing kit for verifying industrial PLC signals in the field.

Electronics Head of Department · UTM Robocon — Jan 2024 – Apr 2025
Malaysia's team at ABU Robocon 2024 and 2025

- Designed 4-layer STM32F4 main-control and STM32G4 motor-interface boards in Altium, with CAN, I^2C and SPI links across all subsystems.
- Upgraded the mainboard's short-circuit protection and layout; designed a custom IMU board for navigation and STM32 + PID shooter control.
- Sole electronics lead between the mechanical and software teams; trained 50+ apprentices in Altium and STM32.

EDUCATION

Bachelor of Electrical Engineering (Hons) · Universiti Teknologi Malaysia — Mar 2023 – Feb 2027
Final-year project: supraharmonic emissions in PV inverter systems, simulated in MATLAB/Simulink · Student member, Saudi Council of Engineers
Certificate: PCB Basic Design (Altium Fundamentals), Altium Education, Sep 2026

PROJECTS

Smart Posture Correction System · Capstone
GOLD MEDAL, MIIE 2026 · INDUSTRY AWARD, UTM EECS 2026

- Python/OpenCV and dlib face-pose pipeline fused with IMU data to classify posture; haptic feedback in under 200 ms.
- 4-layer KiCad PCB (XIAO ESP32-C3, MPU6050), C++ firmware over UDP with Blynk IoT, SolidWorks enclosure.

Datasheet lookup tool — 2026

- Local Python CLI for component datasheets: PyMuPDF extraction, SQLite FTS5 search, incremental indexing, IC tagging.

Other builds

- Self-balancing robot (ESP32, MPU6050, PID) · STM32 line follower with Bluetooth override · custom boards: isolated valve drivers, DC-DC bucks, ESP32 and STM32 dev boards.

LEADERSHIP

- Organising committee and referee, IYRC 2025 (International Youth Robot Competition), Johor · Jun 2025
- Presenter and mentor, month-long Self-Balancing Robot Workshop: PCB design, embedded systems, PID · May 2025
- Mentor, Team Sudan, FIRST Global Challenge 2026, Incheon · Oct 2026
- Workshop instructor, EEExplore, remote · Sep 2025 – now

SKILLS

HARDWARE DESIGN	Altium Designer, KiCad, 4-layer PCB, rigid-flex to IPC-2223, schematic review, SolidWorks
SIMULATION	LTspice, PSpice, ngspice, MATLAB/Simulink, NI Multisim
FIRMWARE	Embedded C on STM32 (CubeMX, CubeIDE), peripheral bring-up and drivers, ESP32, CAN, I^2C, SPI, RS-485
SOFTWARE	Python, C/C++, OpenCV, SQLite
INDUSTRIAL	PLC testing, industrial wiring, DI/DO signal verification